

# Scan-to-Drive

## QR-Based Vehicle Trip Authorization Web App

### Detailed Project Plan

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| Item     | Detail                                                                             |
|----------|------------------------------------------------------------------------------------|
| Document | Detailed project plan: business flow, screens, admin dashboard, technology, server |
| Version  | 1.0 — draft for review                                                             |
| Date     | 5 October 2026                                                                     |

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## 1. Summary

We will build **Scan-to-Drive**, a web app that controls and records every trip of our company cars. Each car has its own QR code. The driver scans the car QR to start a trip, enters the trip details and takes a photo of the dashboard. The passenger then confirms the **start** and the **end** of the trip by scanning a one-time QR code on the driver's phone and entering only his or her employee ID. The admin sees every car and every trip live on a dashboard and gets ready reports.

### Key points

- Works in the browser of any phone, tablet or computer. Nothing to install.
- Every trip has proof: start and end km with dashboard photos, times, places, and the passenger's confirmation.
- The passenger confirms on his own phone, so a driver cannot create or close a trip alone.
- Trips with or without a passenger are both supported.
- Admin dashboard shows live car status, live trips, trip history with photos, alerts and reports (Excel / PDF).
- Built with Next.js (React) for the web app, FastAPI (Python) for the backend and PostgreSQL for the database. Hosted on one Linux VM on Microsoft Azure.

## 2. Why we need it

When car use is tracked by hand, it is hard to know who used which car, how far it really went, and whether the trip really happened. Scan-to-Drive solves this with simple steps that leave proof at every point.

| Problem                                        | How Scan-to-Drive solves it                                                                                                        |
|------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| No clear record of who used which car and when | Every trip is linked to one car, one driver and one passenger, with times taken from the server (not from the phone).              |
| Km readings can be wrong or made up            | Dashboard photo at start and end. Start km is checked against the car's last end km. Admin can see the photo next to the typed km. |
| Trips can be written without a real passenger  | The passenger confirms start and end on his own phone with his employee ID.                                                        |
| Management cannot see where cars are right now | Live car status board: on trip, available, waiting, maintenance.                                                                   |
| Monthly car reports take a lot of manual work  | Reports by car, driver, passenger and department are ready any time, with Excel / PDF export.                                      |
| Problems are found too late                    | Automatic alerts: km gap, long trip, trip not confirmed, wrong IDs.                                                                |

## 3. Scope

### 3.1 In the first version

- Car list and a printable QR sticker for every car.

- Driver flow: scan car QR, start form, dashboard photo, Start QR, end form, end photo, End QR.
- Passenger flow: scan QR, check details, confirm start and end with employee ID.
- Trips with a passenger and trips without a passenger.
- Admin dashboard: live car status, live trips, trip history with photos, alerts, reports, Excel / PDF export.
- Management of cars, drivers, passengers and admin users.
- Business rules, audit log and alerts.
- Bangla and English screens (to confirm).

### 3.2 Not in the first version (can be added later)

- Live GPS tracking during the trip.
- Fuel and maintenance records.
- AI reading of the odometer from the dashboard photo.
- SMS / WhatsApp messages.
- Link with HR or ERP systems.

These are described in Section 20.

## 4. Users and roles

| Role             | Who                                | Main work                                                                          | How they sign in                                    |
|------------------|------------------------------------|------------------------------------------------------------------------------------|-----------------------------------------------------|
| <b>Driver</b>    | Company drivers                    | Start a trip, take dashboard photos, show the QR, end the trip                     | Employee ID + PIN, once per phone (stays signed in) |
| <b>Passenger</b> | Any employee who uses the car      | Confirm the start and the end of the trip                                          | No sign-in. Only enters employee ID on the QR page  |
| <b>Admin</b>     | Transport manager / transport team | Manage cars and people, watch live trips, close stuck trips, check alerts, reports | Username + password                                 |
| <b>Viewer</b>    | Management                         | See the dashboard and reports only                                                 | Username + password                                 |

## 4.1 Who can do what

| Action                                      | Driver  | Passenger | Admin | Viewer |
|---------------------------------------------|---------|-----------|-------|--------|
| Start and end a trip                        | ✓       | –         | –     | –      |
| Confirm trip start and end                  | –       | ✓         | –     | –      |
| Cancel a trip before the passenger confirms | ✓ (own) | –         | ✓     | –      |
| See trip history                            | ✓ (own) | –         | ✓     | ✓      |
| See live dashboard, photos, alerts          | –       | –         | ✓     | ✓      |
| Add / edit cars, drivers, passengers        | –       | –         | ✓     | –      |
| Print car QR stickers                       | –       | –         | ✓     | –      |
| Close a stuck trip (with reason)            | –       | –         | ✓     | –      |
| Export reports (Excel / PDF)                | –       | –         | ✓     | ✓      |
| See audit log and change settings           | –       | –         | ✓     | –      |

## 5. How it works — overview

There are three kinds of users and one web app. Drivers and passengers use their phones. The admin uses a PC or laptop. All of them open the same web app in a browser. The web app talks to the backend server, which checks the rules and saves everything in the database. Dashboard photos are kept in photo storage.

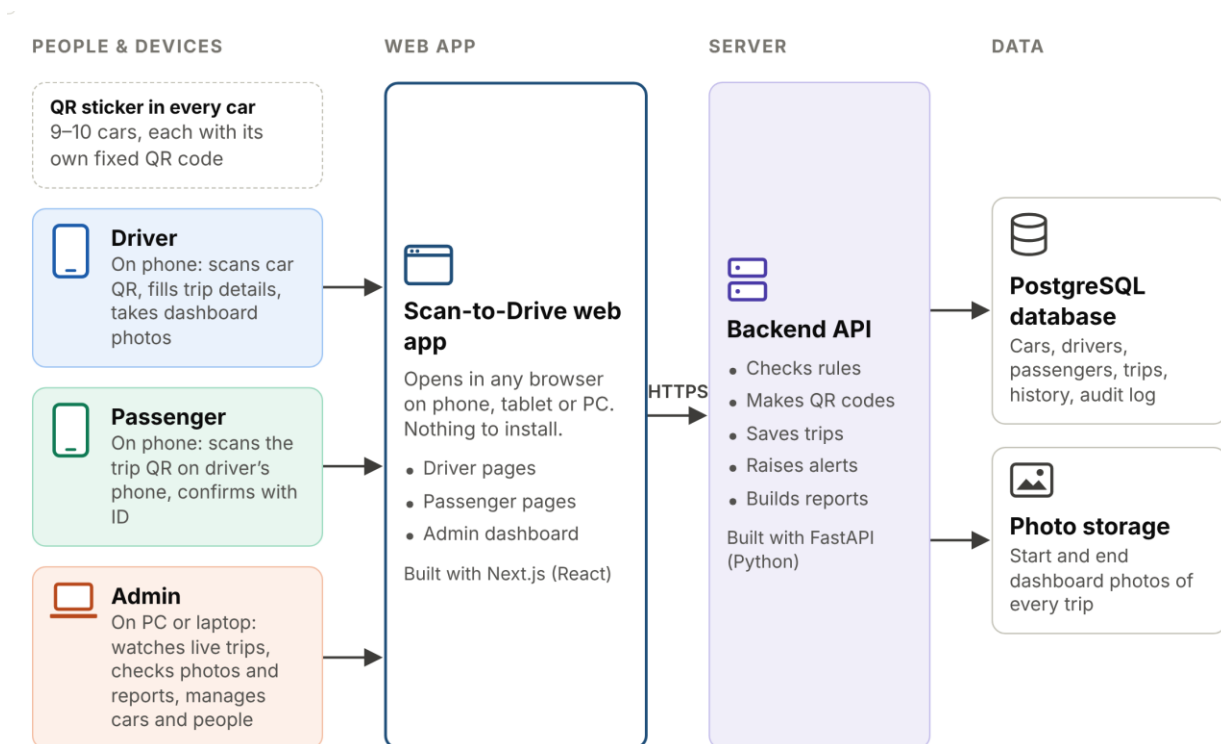


Figure 1: How the parts connect

## 5.1 The trip in 4 steps

Every trip with a passenger follows the same 4 steps. The trip status changes after each step, and the admin dashboard shows it live.

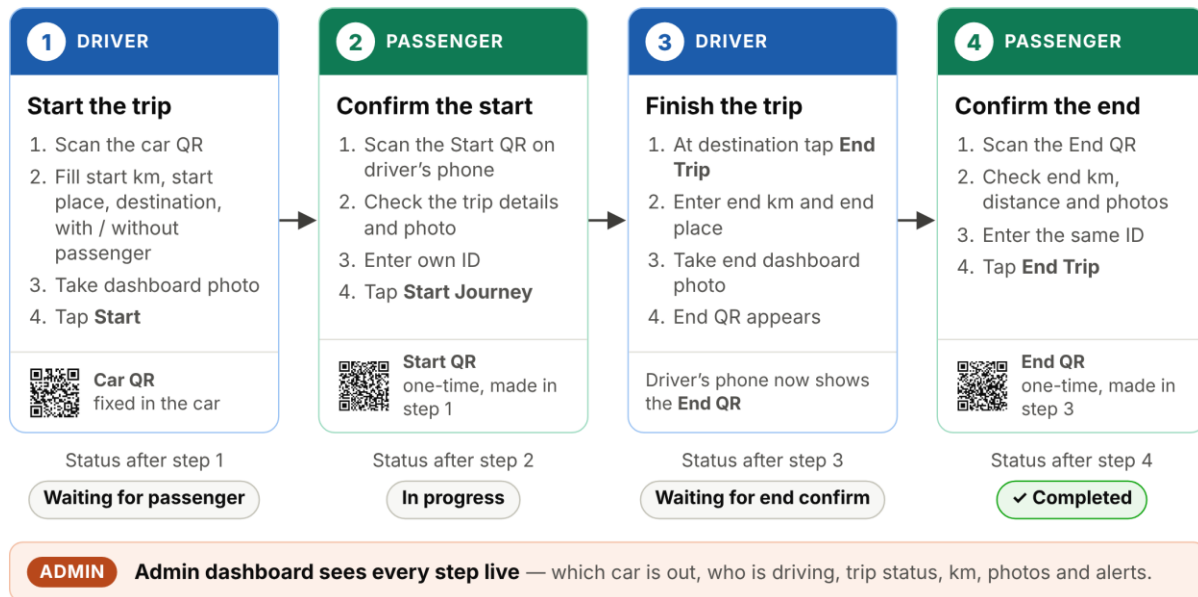


Figure 2: The trip in 4 steps

**Why the End QR comes after the end details:** the driver first types the end km and takes the end photo, then the End QR appears. This way the passenger checks the real end km and photo before confirming.

## 6. Detailed business flow

### 6.1 One-time setup by admin

Before the first trip, the admin sets up the system once. After that, only new cars or new people need to be added.

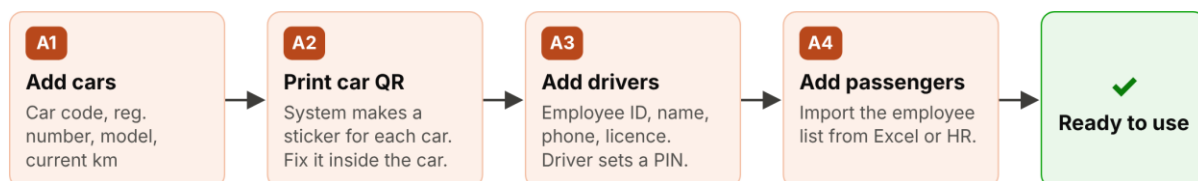


Figure 3: One-time setup by admin

- Add cars:** car code (CAR-01 ... CAR-10), registration number, model and current odometer km.
- Print car QR:** the system makes a QR sticker for each car (PDF to print). The sticker is fixed inside the car, for example on the dashboard or sun visor. Laminated stickers last longer.
- Add drivers:** employee ID, name, phone and licence number. Each driver sets a 4-digit PIN on first sign-in.
- Add passengers:** import the employee list (ID, name, department) from Excel or from HR.

## 6.2 Part A — Start of the trip and passenger pickup

The picture below shows who does what, step by step. Blue = driver, purple = system, green = passenger.

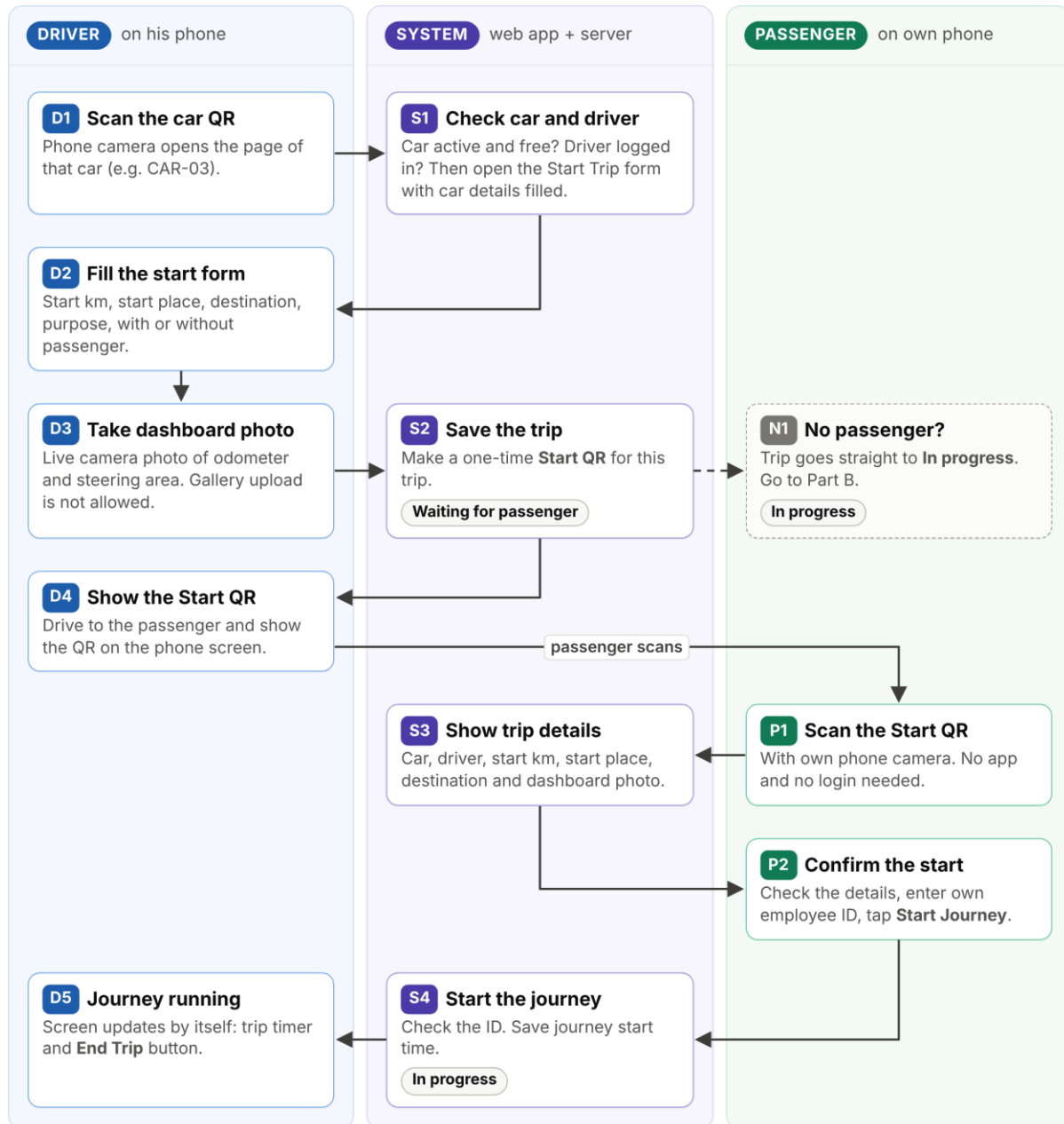


Figure 4: Detailed flow, Part A — start of the trip

| Step | What happens              | Details                                                                                                                     | System checks                                          |
|------|---------------------------|-----------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| D1   | Driver scans the car QR   | Normal phone camera opens the page of that car.                                                                             | Car exists and is active.                              |
| S1   | System opens the car page | If the driver is not signed in, he signs in with employee ID + PIN. Then the Start Trip form opens with car details filled. | Car is not in maintenance. Car has no other open trip. |

| Step      | What happens                 | Details                                                                                           | System checks                                                                           |
|-----------|------------------------------|---------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| <b>D2</b> | Driver fills the start form  | Start km, start place (GPS helps), destination, purpose, with or without passenger.               | Start km is not lower than the car's last end km. Big gap → alert.                      |
| <b>D3</b> | Driver takes dashboard photo | Live camera photo of the odometer and steering area.                                              | Photo is required. Gallery photos are not accepted.                                     |
| <b>S2</b> | System saves the trip        | Trip number is created (e.g. T-0458). Status: Waiting for passenger. A one-time Start QR is made. | Server time is saved. Car is held for this trip.                                        |
| <b>D4</b> | Driver shows the Start QR    | Driver goes to the passenger and shows the QR on his phone.                                       | Driver can make a new QR or cancel the trip.                                            |
| <b>P1</b> | Passenger scans the Start QR | With his own phone camera. No app, no sign-in.                                                    | QR is valid, not used, not expired.                                                     |
| <b>S3</b> | System shows trip details    | Car, driver, start km, start place, destination and dashboard photo.                              | —                                                                                       |
| <b>P2</b> | Passenger confirms the start | Enters own employee ID and taps Start Journey. His name is shown to confirm.                      | ID exists and is active. Not sent from the driver's own phone. Wrong tries are limited. |
| <b>S4</b> | System starts the journey    | Status: In progress. Journey start time is saved. Driver's screen updates by itself.              | Start QR can't be used again. Event is logged.                                          |
| <b>D5</b> | Journey running              | Driver's screen shows a timer and the End Trip button.                                            | —                                                                                       |

### 6.3 Part B — End of the trip



Figure 5: Detailed flow, Part B — end of the trip

| Step | What happens                   | Details                                                         | System checks                                            |
|------|--------------------------------|-----------------------------------------------------------------|----------------------------------------------------------|
| D6   | Driver reaches the destination | Opens the running trip and taps End Trip.                       | Only the driver of this trip can end it.                 |
| D7   | Driver fills the end form      | End km, end place (GPS helps) and a new live dashboard photo.   | End km must be more than start km. Very high km → alert. |
| S5   | System checks and saves        | Status: Waiting for end confirm. A new one-time End QR is made. | Distance and time are calculated.                        |
| D8   | Driver shows the End QR        | Shows the QR on his phone to the passenger.                     | QR can be refreshed if it expires.                       |



| Step | What happens               | Details                                                                           | System checks                                  |
|------|----------------------------|-----------------------------------------------------------------------------------|------------------------------------------------|
| P3   | Passenger scans the End QR | With his own phone camera.                                                        | QR is valid, not used, not expired.            |
| S6   | System shows trip summary  | Start and end km, distance, time taken, both photos.                              | —                                              |
| P4   | Passenger confirms the end | Enters the same employee ID and taps End Trip.                                    | ID must match the ID used at the start.        |
| S7   | System closes the trip     | Status: Completed. Trip is locked. Car becomes free. Car's current km is updated. | Event is logged. Dashboard and reports update. |
| D9   | Trip completed             | Driver sees the summary. Car is ready for the next trip.                          | —                                              |

## 6.4 Trips without a passenger

Sometimes the driver drives alone (for example to fuel the car, go to the workshop or pick up goods). In this case:

- The driver selects **Without passenger** in the start form.
- After the start dashboard photo, the trip goes straight to **In progress**. No Start QR is shown.
- At the end, the driver enters end km, end place and the end photo, and the trip becomes **Completed**. No End QR is shown.
- These trips are marked “No passenger” in reports, so the admin can review them. The purpose field is required for them.

**To confirm:** whether no-passenger trips also need approval by the admin or a supervisor. See Section 21.

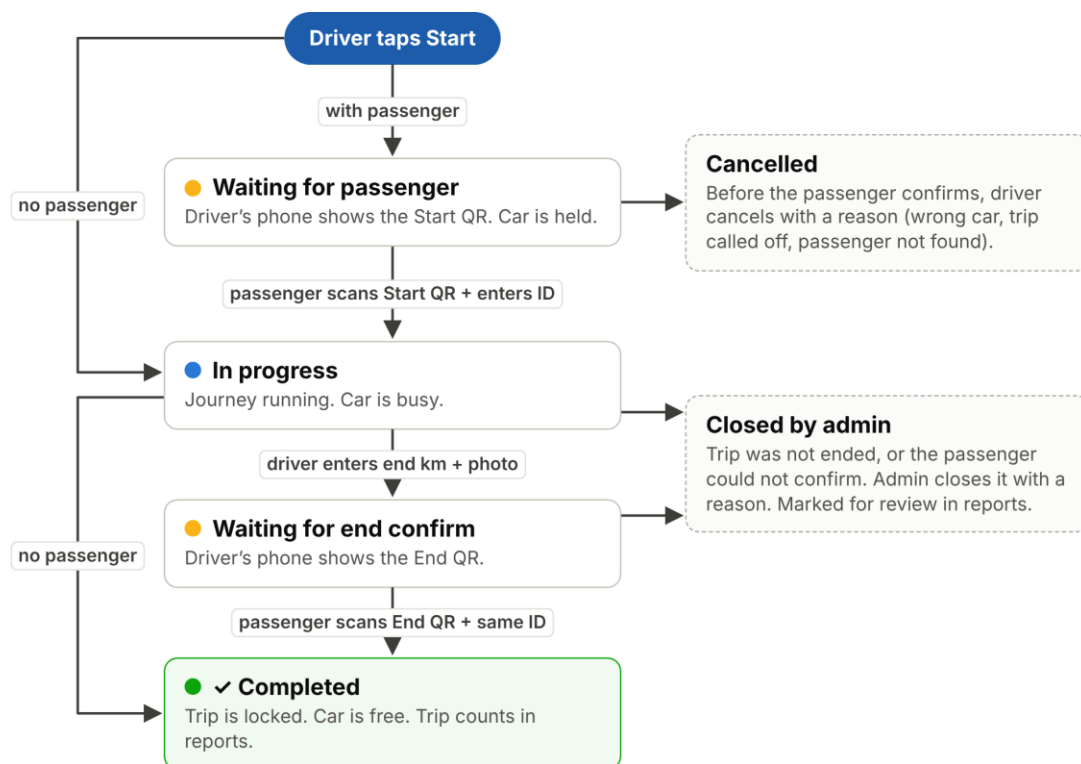
## 6.5 Special cases

| Situation                                    | What the system does                                                                                            |
|----------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| Car already has an open trip                 | Driver sees “CAR-03 is in use by Karim H.” and cannot start. Admin can close the old trip with a reason.        |
| Car is in maintenance                        | Trip cannot start. Message shown to the driver.                                                                 |
| Start km is lower than the last end km       | Blocked. Driver must check and correct the km.                                                                  |
| Start km is much higher than the last end km | Allowed, but a “Km gap” alert goes to the admin to check the photos (car may have been used without a trip).    |
| Start QR or End QR expired                   | Driver taps “Make new QR”. The old QR stops working.                                                            |
| Passenger not found or trip called off       | Driver cancels the trip with a reason before the passenger confirms. Car becomes free. Trip shows as Cancelled. |
| Wrong passenger ID                           | “ID not found” message. After 5 wrong tries the QR is blocked and the admin gets an alert.                      |
| A different person tries to end the trip     | Rejected. The end must be confirmed with the same ID as the start.                                              |

| Situation                                   | What the system does                                                                                                                                  |
|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| Passenger has no phone or no internet       | Passenger can scan with any other phone. If not possible, driver taps “Passenger can’t scan” and the admin approves later. Trip is marked for review. |
| Driver forgets to end the trip              | Reminder to the driver after a set number of hours. Alert to the admin. Admin can close the trip with a reason.                                       |
| Weak network while uploading the photo      | Photo stays on the phone and upload is retried. The step is not saved until the photo is uploaded.                                                    |
| Driver scanned the wrong car                | Driver cancels before the passenger confirms, then scans the right car.                                                                               |
| Driver’s phone switches off during the trip | Driver signs in on another phone. The open trip appears again and he can end it.                                                                      |

## 7. Trip status

Each trip moves through clear statuses. The status decides what each screen shows and whether the car is free.



**No passenger trips** skip the passenger steps: the driver starts and ends the trip with dashboard photos only, and the admin can review them.

Figure 6: Trip status and how it changes

| Status                  | Meaning                                                                    | Car  | Shown to driver  |
|-------------------------|----------------------------------------------------------------------------|------|------------------|
| Waiting for passenger   | Driver started; passenger has not confirmed yet                            | Held | Start QR         |
| In progress             | Journey is running                                                         | Busy | Timer + End Trip |
| Waiting for end confirm | Driver ended; passenger has not confirmed yet                              | Busy | End QR           |
| Completed               | Passenger confirmed the end (or driver ended a no-passenger trip). Locked. | Free | Summary          |
| Cancelled               | Stopped before the passenger confirmed the start                           | Free | Reason           |
| Closed by admin         | Stuck trip closed by the admin with a reason; marked for review            | Free | Reason           |

## 8. QR codes

There are three kinds of QR codes. All of them are simple web links, so they open with the **normal phone camera**. No QR app is needed. The web app also has a “Scan” button as a backup for old phones.

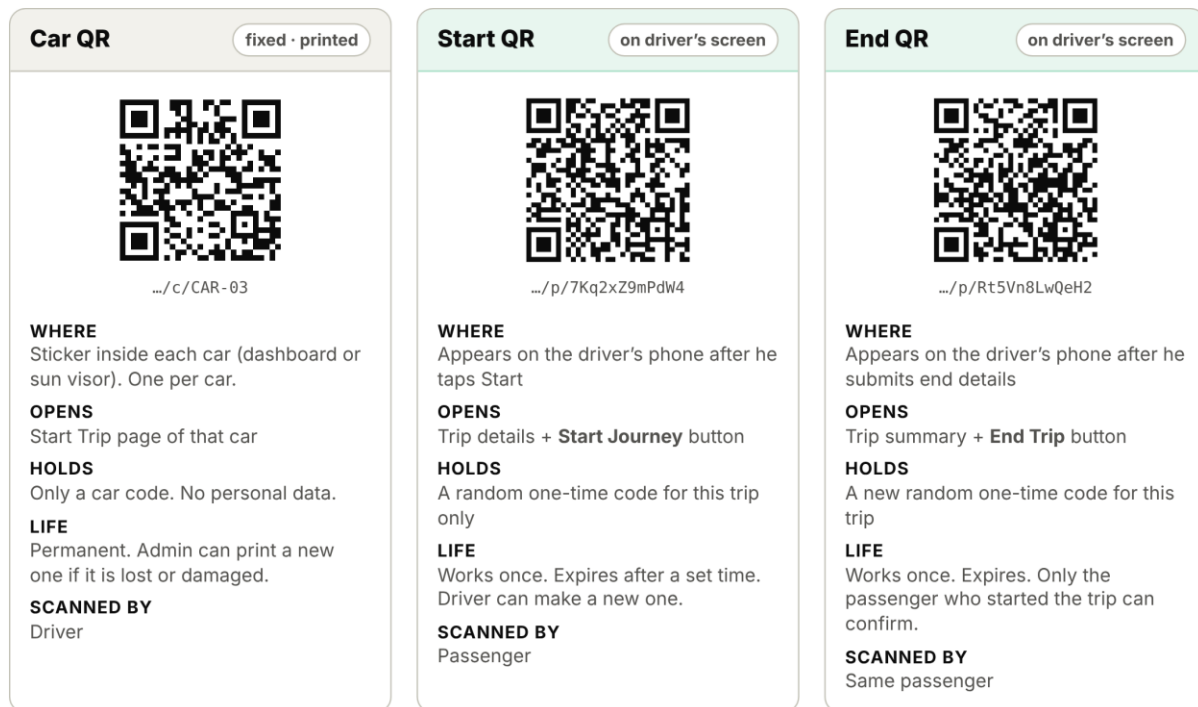


Figure 7: The three QR codes

### Rules that keep the QR codes safe

- The car QR holds only a car code. It does not hold any personal data.
- Start and End QR codes hold a long random code that is **used once** and **expires** after a set time (for example 15 minutes; admin can change it).

- Only a hidden form (hash) of the code is kept in the database, so a leaked database cannot be used to make QR codes.
- A screenshot of an old QR does not work, because each code works only once.
- If a car sticker is lost, the admin prints a new one. The old sticker stops working (QR version number).
- Sticker: about 6 × 6 cm, laminated, with the car code printed under the QR (for example “CAR-03 — scan to start trip”).

## 9. Information we capture

| Stage                | Field                                | Given by  | How                                         |
|----------------------|--------------------------------------|-----------|---------------------------------------------|
| <b>Trip start</b>    | Car                                  | System    | From the car QR                             |
|                      | Driver name and ID                   | System    | From driver sign-in                         |
|                      | Start km                             | Driver    | Typed (last end km shown as a hint)         |
|                      | Start place                          | Driver    | Typed or picked; GPS point saved if allowed |
|                      | Destination and purpose              | Driver    | Typed or picked from a list                 |
|                      | With / without passenger             | Driver    | Selected                                    |
|                      | Start dashboard photo                | Driver    | Live camera                                 |
|                      | Start time                           | System    | Server time                                 |
| <b>Start confirm</b> | Passenger ID (and name)              | Passenger | Typed; name shown back to confirm           |
|                      | Journey start time                   | System    | Server time                                 |
| <b>Trip end</b>      | End km                               | Driver    | Typed (start km shown)                      |
|                      | End place                            | Driver    | Typed or picked; GPS point saved if allowed |
|                      | End dashboard photo                  | Driver    | Live camera                                 |
|                      | End time                             | System    | Server time                                 |
| <b>End confirm</b>   | Passenger ID                         | Passenger | Must match the start ID                     |
|                      | End confirm time                     | System    | Server time                                 |
| <b>Calculated</b>    | Distance, journey time, waiting time | System    | End km – start km; time differences         |
| <b>Every step</b>    | Audit record                         | System    | Who, what, when, which device, IP, GPS      |

## 10. Screens

The screens below are early sketches to show the idea. Final design will follow the company look. Names and numbers are sample data.

### 10.1 Driver screens

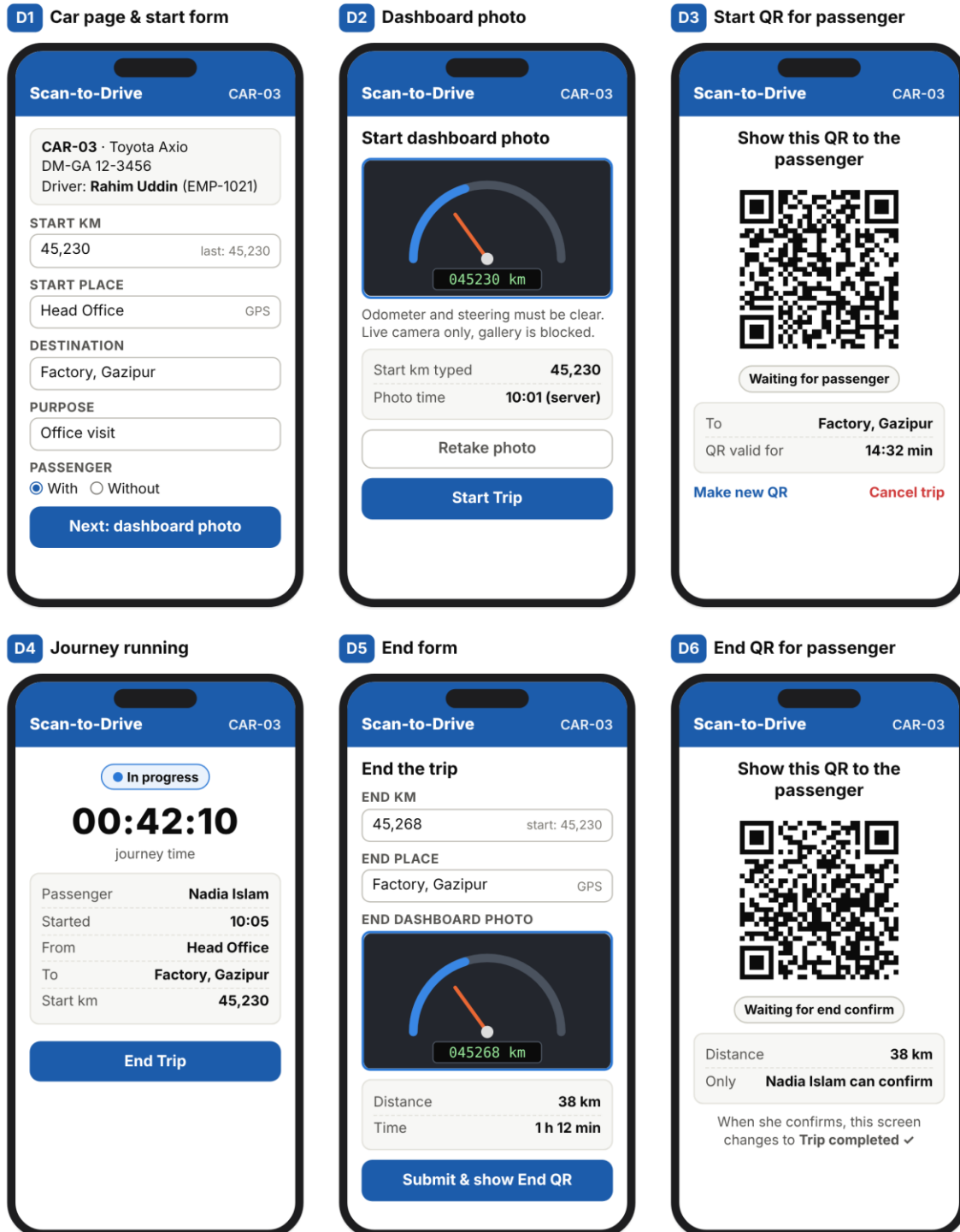


Figure 8: Driver screens (sketch)

## 10.2 Passenger screens

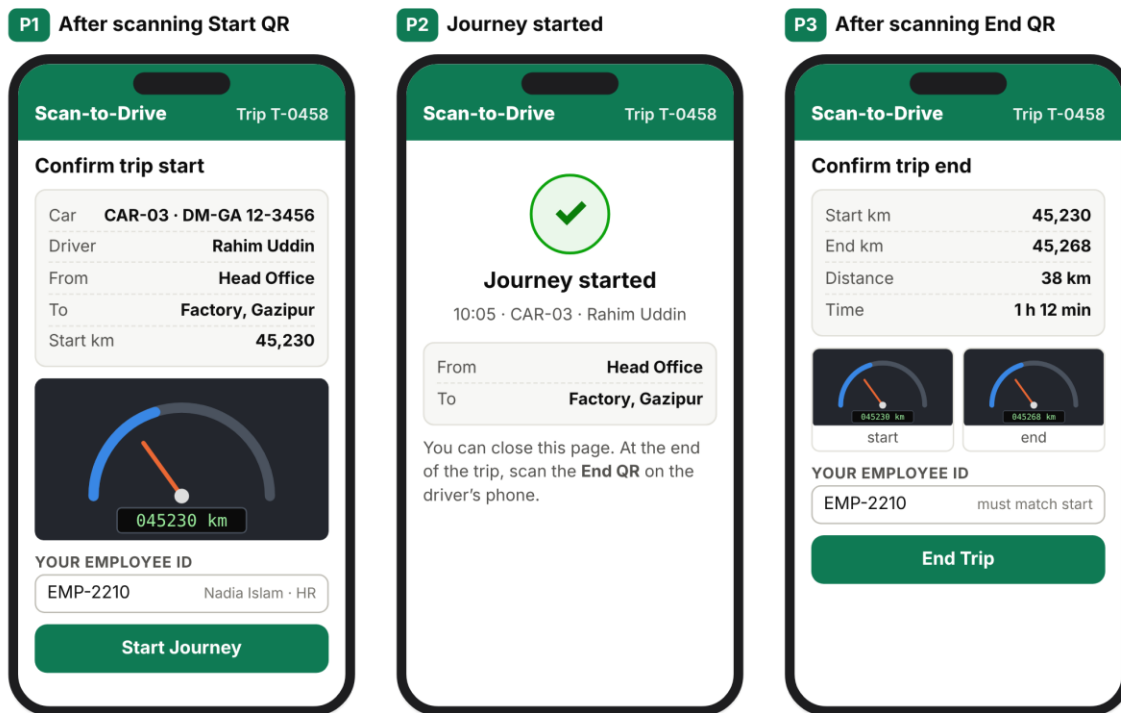


Figure 9: Passenger screens (sketch)

## 10.3 Screen design rules

- Mobile first: big buttons, large text, one main action per screen.
- Bangla and English switch on every screen (to confirm).
- Works on normal Android phones (Chrome) and iPhones (Safari).
- Photos are made smaller on the phone before upload, to save data and time.
- Driver screen updates by itself when the passenger confirms. No need to refresh.
- Clear messages for every error, for example “QR expired — ask the driver to make a new one”.
- Status is always shown with text and colour together, never colour alone.

## 11. Admin dashboard

The admin dashboard is for the transport team and management. It opens on a PC or laptop (it also works on a tablet). It shows what is happening right now and what happened before.

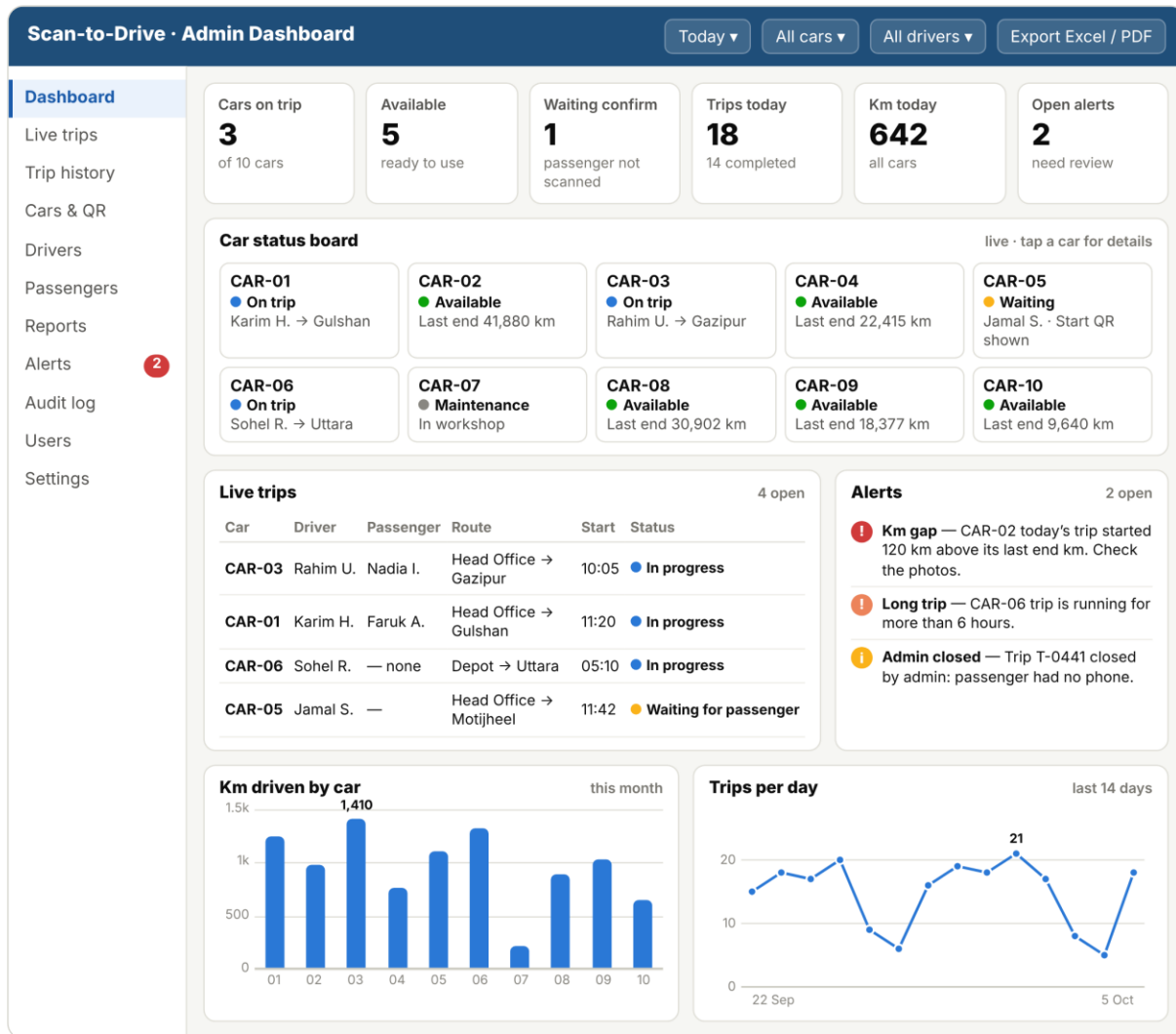


Figure 10: Admin dashboard home page (sketch with sample data)

## 11.1 What the home page shows

| Part             | What it shows                                                                                                                                              |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Summary cards    | Cars on trip, available cars, trips waiting for confirm, trips today, km today, open alerts.                                                               |
| Car status board | All 9–10 cars as tiles: On trip (with driver and destination), Available (with last km), Waiting, Maintenance. Click a car to see its details and history. |
| Live trips       | Every open trip: car, driver, passenger, route, start time, status. Click a trip to see photos and details.                                                |
| Alerts           | Problems that need a look: km gap, long trip, wrong IDs, admin-closed trips.                                                                               |
| Charts           | Km driven by each car this month; trips per day; more charts on the Reports page.                                                                          |
| Filters          | Date range (today, this week, this month, custom), car, driver, department. Export to Excel or PDF.                                                        |

## 11.2 Admin pages

| Page                 | What it shows                                                                | What the admin can do                                                                                             |
|----------------------|------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| <b>Live trips</b>    | All open trips with status and time running                                  | Open a trip, call the driver, close a stuck trip with a reason                                                    |
| <b>Trip history</b>  | All past trips with filters and search                                       | Open trip detail: timeline of every step, start and end photos side by side, km, places, map points, audit record |
| <b>Cars &amp; QR</b> | Car list, status, current km, total trips                                    | Add / edit car, set maintenance, print or re-print QR sticker                                                     |
| <b>Drivers</b>       | Driver list, status, trips and km                                            | Add / edit driver, reset PIN, deactivate                                                                          |
| <b>Passengers</b>    | Employee list with department                                                | Import from Excel, add / edit, deactivate                                                                         |
| <b>Reports</b>       | Ready reports (see 11.3)                                                     | Filter, view charts, export Excel / PDF                                                                           |
| <b>Alerts</b>        | Open and closed alerts                                                       | Check, add a note, mark as solved                                                                                 |
| <b>Audit log</b>     | Every action in the system, who and when                                     | Search and export (read only)                                                                                     |
| <b>Users</b>         | Admin and viewer accounts                                                    | Add users, set role, reset password                                                                               |
| <b>Settings</b>      | QR expiry time, km gap limit, long-trip hours, wrong-ID limit, purposes list | Change settings (changes are logged)                                                                              |

## 11.3 Reports

| Report                        | What it shows                                                                               |
|-------------------------------|---------------------------------------------------------------------------------------------|
| <b>Car usage</b>              | Per car: number of trips, total km, hours used, idle days, km per trip.                     |
| <b>Driver report</b>          | Per driver: trips, km, average trip time, cancelled trips, alerts.                          |
| <b>Passenger / department</b> | Trips and km by passenger and by department (useful for cost sharing).                      |
| <b>Trip log</b>               | Every trip with all fields and links to photos.                                             |
| <b>Exceptions</b>             | Cancelled trips, admin-closed trips, km gaps, long trips, no-passenger trips.               |
| <b>Km continuity</b>          | Per car: each trip's start km against the previous end km, to find unrecorded use.          |
| <b>Monthly summary</b>        | One page for management: total trips, total km, busiest cars, top departments, open issues. |

All reports can be filtered by date, car, driver and department, and exported to Excel or PDF.



## 11.4 Alerts

| Alert                   | When it is raised                                                        | Goes to                     |
|-------------------------|--------------------------------------------------------------------------|-----------------------------|
| <b>Km gap</b>           | Start km is higher than the car's last end km by more than the set limit | Admin                       |
| <b>Long trip</b>        | Trip is In progress for longer than the set hours                        | Admin + reminder to driver  |
| <b>Waiting too long</b> | Trip stays Waiting for passenger or Waiting for end confirm too long     | Admin + reminder to driver  |
| <b>Wrong IDs</b>        | Too many wrong passenger IDs on one trip                                 | Admin                       |
| <b>High km</b>          | Trip distance is far above normal for that route or car                  | Admin                       |
| <b>Admin closed</b>     | A trip was closed by the admin                                           | Shown in reports for review |

## 12. Business rules

1. One car can have only **one open trip** at a time.
2. A car in **maintenance** or **inactive** cannot start a trip.
3. A driver can have only one open trip at a time.
4. **Dashboard photo** is required at the start and at the end. Only live camera photos are accepted.
5. Start km must **not be lower** than the car's last end km. A gap above the set limit raises an alert.
6. End km must be **higher** than start km.
7. All times are **server times**, not phone times.
8. The passenger who confirms the end must be the **same** passenger who confirmed the start.
9. Passenger confirmation is **not accepted from the driver's own signed-in phone**.
10. Start and End QR codes work **once** and **expire** after a set time.
11. A driver can cancel a trip **only before** the passenger confirms the start. After that only the admin can close it.
12. Completed trips are **locked**. Only the admin can correct a trip, and every change is saved in the audit log with a reason.
13. Cars, drivers and passengers are **deactivated**, never deleted, so old trips stay correct.

## 13. Security and anti-misuse

| Risk                                                                                | How we stop it                                                                                                                            |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Driver types wrong km</b>                                                        | Dashboard photos at start and end; km checked against last trip; passenger sees the km and photo; admin sees photos next to typed km.     |
| <b>Driver starts a trip without being in the car (using an old photo of the QR)</b> | Live camera photo of the dashboard is required; GPS point is saved when allowed; server time is used.                                     |
| <b>Driver confirms for the passenger</b>                                            | Confirmation is blocked from the driver's signed-in phone; passenger name is shown back; option to add a passenger PIN or SMS code later. |

| Risk                         | How we stop it                                                                                                      |
|------------------------------|---------------------------------------------------------------------------------------------------------------------|
| Old QR screenshot is reused  | Start and End QR codes work once and expire.                                                                        |
| Someone guesses IDs          | Limit on wrong tries; QR blocked after the limit; alert to admin.                                                   |
| Data is changed later        | Completed trips are locked; admin changes need a reason and are logged; audit log is read only.                     |
| Unauthorized access          | HTTPS only; PINs and passwords stored as secure hashes; role-based access; sessions expire; rate limits on sign-in. |
| Photos are seen by outsiders | Photos are in private storage; only short-lived links are given to signed-in admins and the trip's passenger page.  |
| Server attack                | Firewall allows only web ports; SSH only from the office IP; automatic security updates; daily backups.             |

## 14. Technology stack

The stack is the one we agreed: **Next.js (React)** for the web app, **FastAPI** for the backend and **PostgreSQL** for the database. Below are the main parts and why we use them.

### 14.1 Web app (frontend)

| Part                    | Technology                                          | Why                                                                    |
|-------------------------|-----------------------------------------------------|------------------------------------------------------------------------|
| Framework               | Next.js (React, TypeScript)                         | One web app for driver, passenger and admin; fast; runs in any browser |
| Look and layout         | Tailwind CSS + shadcn/ui                            | Clean, mobile-first screens that are quick to build                    |
| Forms                   | React Hook Form + Zod                               | Easy forms with checks before sending                                  |
| Data and live updates   | TanStack Query, short polling or Server-Sent Events | Driver screen updates by itself when the passenger confirms            |
| Show QR on screen       | qrcode.react                                        | Draws the Start and End QR on the driver's phone                       |
| Scan QR in app (backup) | html5-qrcode                                        | For phones whose camera app can't read QR codes                        |
| Camera photo            | Browser camera + image compression                  | Live photo only; smaller files for weak networks                       |
| Charts                  | Recharts                                            | Charts on the admin dashboard                                          |
| Languages               | next-intl                                           | Bangla and English                                                     |
| Install as app          | PWA (Progressive Web App)                           | Drivers can add an icon to the home screen                             |

### 14.2 Backend

| Part          | Technology            | Why                                       |
|---------------|-----------------------|-------------------------------------------|
| API framework | FastAPI (Python 3.12) | Fast, simple, automatic API documentation |

| Part                  | Technology                                      | Why                                            |
|-----------------------|-------------------------------------------------|------------------------------------------------|
| Database access       | SQLAlchemy 2 + Alembic                          | Safe database code and versioned table changes |
| Data checks           | Pydantic v2                                     | Every request is checked before use            |
| Sign-in               | JWT tokens; bcrypt hashing for PIN and password | Secure sessions for drivers and admins         |
| QR codes and stickers | Python QR library + ReportLab (PDF)             | Makes trip QR codes and printable car stickers |
| Photo storage         | Azure Blob Storage SDK                          | Private photo storage with short-lived links   |
| Background jobs       | APScheduler worker                              | Reminders, alerts, nightly backup              |
| Exports               | openpyxl (Excel), ReportLab (PDF)               | Report downloads                               |
| App server            | Gunicorn + Uvicorn workers                      | Runs the API reliably                          |

### 14.3 Database, server and tools

| Part             | Technology                                 | Why                                                         |
|------------------|--------------------------------------------|-------------------------------------------------------------|
| Database         | PostgreSQL 16                              | Reliable, free, strong for reports                          |
| Web server       | Nginx                                      | HTTPS entry point and reverse proxy                         |
| Packaging        | Docker + Docker Compose                    | Same setup on test and live servers; easy updates           |
| Operating system | Ubuntu Server 24.04 LTS                    | Long-term support, stable                                   |
| SSL certificate  | Let's Encrypt (free, auto renew)           | HTTPS is required — browsers allow the camera only on HTTPS |
| Code and deploy  | Git + GitHub Actions                       | Tested, repeatable deployments                              |
| Monitoring       | Azure Monitor + app logs (Sentry optional) | Know quickly if something breaks                            |

## 15. API design

Main backend endpoints (all under /api). Admin endpoints need an admin sign-in. Passenger endpoints work only with a valid one-time QR code.

| Method | Path                 | Used by | Purpose                             |
|--------|----------------------|---------|-------------------------------------|
| POST   | /auth/driver/login   | Driver  | Sign in with employee ID + PIN      |
| POST   | /auth/admin/login    | Admin   | Sign in with username + password    |
| GET    | /cars/{car_code}     | Driver  | Car page after scanning the car QR  |
| POST   | /trips               | Driver  | Start a trip (form data)            |
| POST   | /trips/{id}/photos   | Driver  | Upload start or end dashboard photo |
| POST   | /trips/{id}/start-qr | Driver  | Make (or re-make) the Start QR      |

| Method | Path                                           | Used by        | Purpose                                           |
|--------|------------------------------------------------|----------------|---------------------------------------------------|
| GET    | /trips/active                                  | Driver         | Driver's open trip and live status                |
| POST   | /trips/{id}/end                                | Driver         | Submit end km, end place, end photo; makes End QR |
| POST   | /trips/{id}/cancel                             | Driver / Admin | Cancel before the passenger confirms              |
| GET    | /p/{token}                                     | Passenger      | Trip details or summary for this QR               |
| POST   | /p/{token}/confirm-start                       | Passenger      | Confirm start with employee ID                    |
| POST   | /p/{token}/confirm-end                         | Passenger      | Confirm end with the same employee ID             |
| GET    | /admin/dashboard                               | Admin / Viewer | Summary cards, car board, live trips              |
| GET    | /admin/trips                                   | Admin / Viewer | Trip history with filters                         |
| GET    | /admin/trips/{id}                              | Admin / Viewer | Trip detail with photos and timeline              |
| POST   | /admin/trips/{id}/close                        | Admin          | Close a stuck trip with a reason                  |
| CRUD   | /admin/cars, /admin/drivers, /admin/passengers | Admin          | Manage cars and people; import passengers         |
| GET    | /admin/cars/{id}/qr.pdf                        | Admin          | Printable QR sticker                              |
| GET    | /admin/reports/{name}                          | Admin / Viewer | Report data; ?format=xlsx or pdf to export        |
| GET    | /admin/alerts                                  | Admin          | List alerts; mark as solved                       |
| GET    | /admin/audit                                   | Admin          | Audit log search                                  |

## 16. Database design

The database has one main table, **trips**, linked to cars, drivers and passengers. Photos, QR codes, events and alerts are kept in their own tables, each linked to a trip.

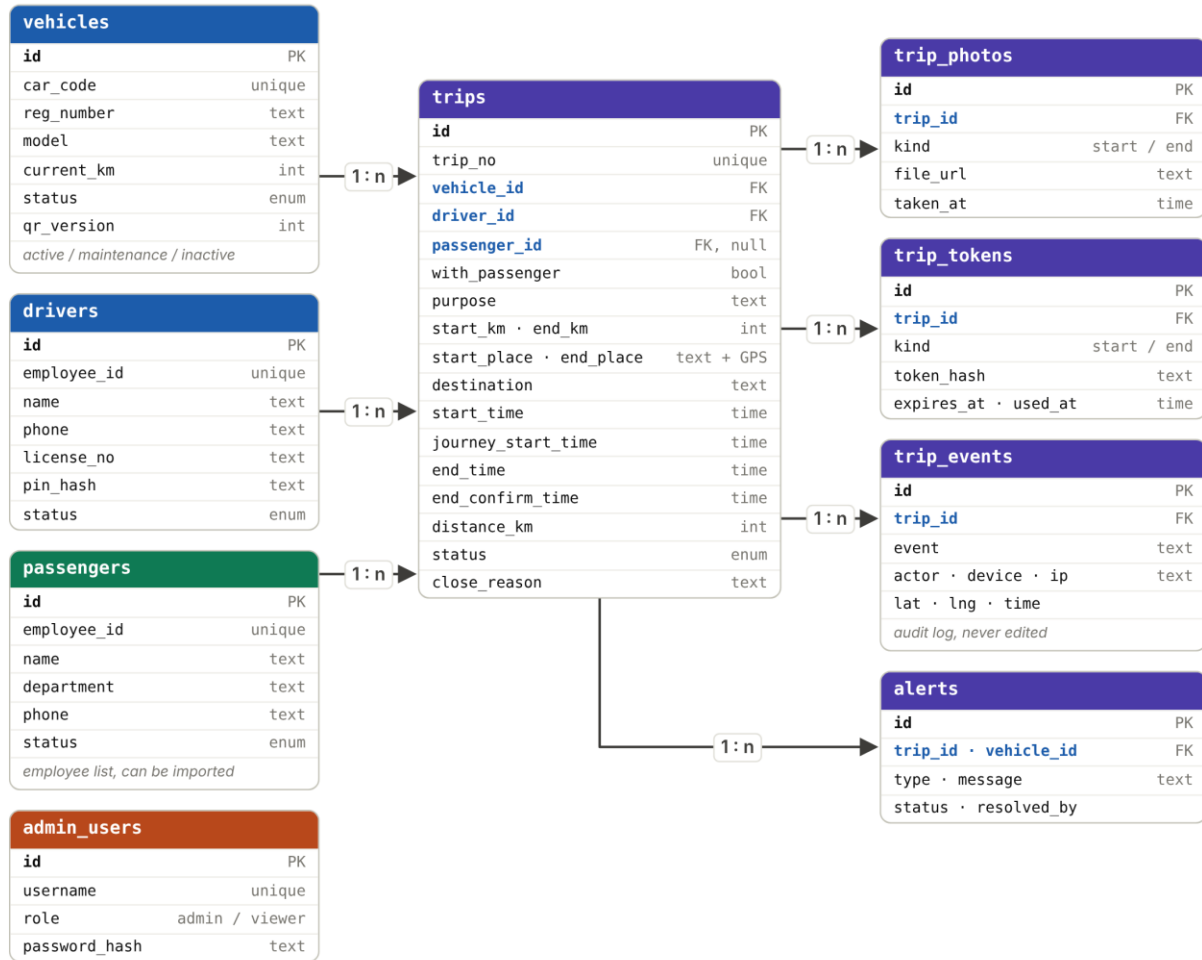


Figure 11: Main database tables and links

| Table              | What it stores                                                                                               |
|--------------------|--------------------------------------------------------------------------------------------------------------|
| <b>vehicles</b>    | Each car: code, registration number, model, current km, status (active / maintenance / inactive), QR version |
| <b>drivers</b>     | Each driver: employee ID, name, phone, licence number, PIN (hashed), status                                  |
| <b>passengers</b>  | Employees who can be passengers: employee ID, name, department, phone, status                                |
| <b>trips</b>       | One row per trip: car, driver, passenger, start and end km, places, times, distance, status, close reason    |
| <b>trip_photos</b> | Start and end dashboard photos: link to the photo file, time taken                                           |
| <b>trip_tokens</b> | One-time Start and End QR codes: hashed code, expiry time, used time                                         |
| <b>trip_events</b> | Audit log for every step: who, what, when, device, IP, GPS. Never edited.                                    |
| <b>alerts</b>      | Alerts for trips or cars: type, message, status, who solved it                                               |
| <b>admin_users</b> | Admin and viewer accounts with roles                                                                         |
| <b>settings</b>    | Values the admin can change: QR expiry, km gap limit, long-trip hours, wrong-ID limit                        |

## Design notes

- Times are saved in UTC and shown in Bangladesh time (UTC+6).
- Photos are saved in Azure Blob Storage; the database keeps only the link.
- Indexes on trip status, car, driver and start time keep the dashboard and reports fast.
- A database rule makes sure one car can't have two open trips at the same time.

## 17. Server and hosting on Azure

For 9–10 cars the load is small, so **one Linux virtual machine (VM)** on Microsoft Azure is enough to run the whole system: web app, API, database and background worker. Photos and backups go to Azure Blob Storage.

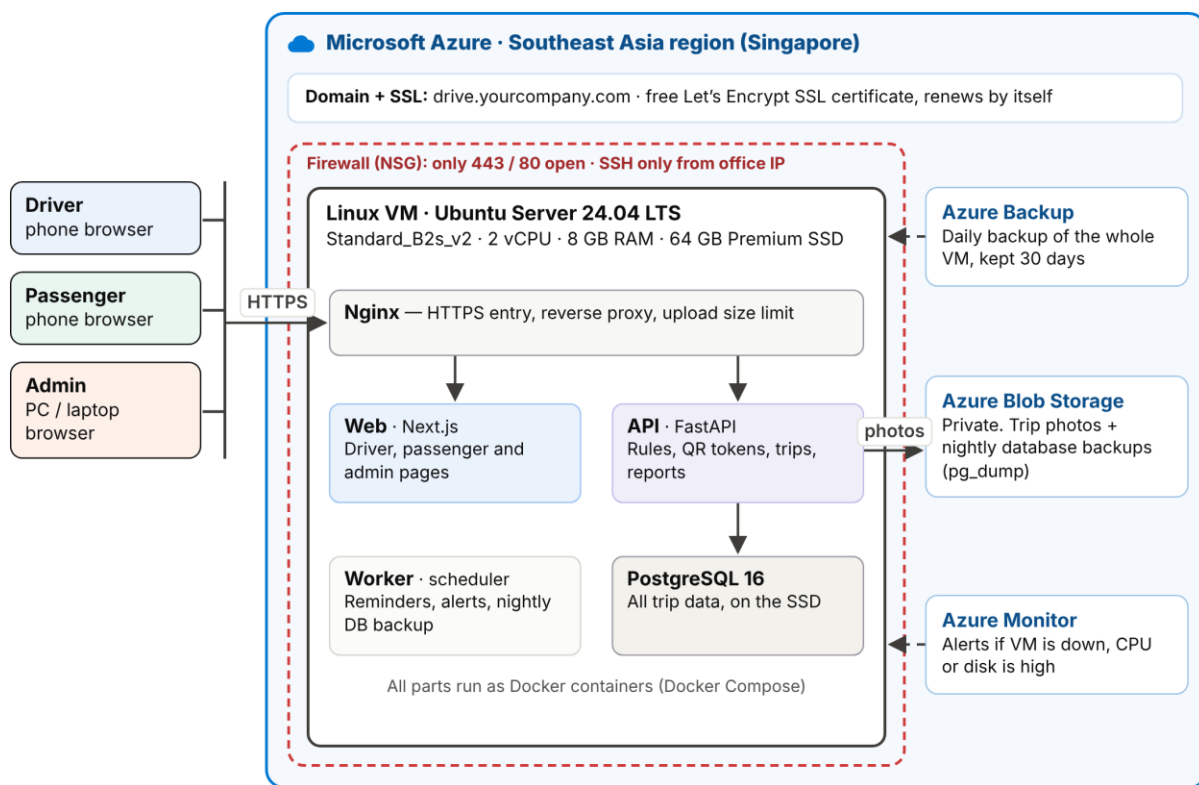


Figure 12: Suggested server setup on Microsoft Azure

### 17.1 Recommended setup

| Item             | Recommendation                                                                        |
|------------------|---------------------------------------------------------------------------------------|
| Cloud            | Microsoft Azure                                                                       |
| Region           | Southeast Asia (Singapore) — close to Bangladesh, low delay                           |
| VM size          | <b>Standard_B2s_v2</b> — 2 vCPU, 8 GB RAM (burstable, good for light and uneven load) |
| Operating system | Ubuntu Server 24.04 LTS                                                               |
| Disk             | 64 GB Premium SSD for system, Docker and database                                     |

| Item                            | Recommendation                                                                                                                                 |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Photo and backup storage</b> | Azure Blob Storage, private container (Hot tier for new photos; old photos can move to the cheaper Cool tier)                                  |
| <b>Network</b>                  | Static public IP. Firewall (Network Security Group): only ports 443 and 80 open; SSH only from the office IP (or Azure Bastion)                |
| <b>Domain and SSL</b>           | A sub-domain such as drive.<company domain> with a free Let's Encrypt certificate that renews by itself. HTTPS is a must for the phone camera. |
| <b>Software</b>                 | Docker + Docker Compose running Nginx, Next.js web app, FastAPI API, PostgreSQL 16 and the worker                                              |
| <b>Backup</b>                   | Azure Backup: daily VM backup kept 30 days. Plus a nightly database dump to Blob Storage kept 30 days.                                         |
| <b>Monitoring</b>               | Azure Monitor alerts when the VM is down, CPU is high or the disk is above 80%                                                                 |

## 17.2 Size options

| Option             | Azure VM size    | CPU / RAM      | When to use                                               |
|--------------------|------------------|----------------|-----------------------------------------------------------|
| <b>Minimum</b>     | Standard_B2ls_v2 | 2 vCPU · 4 GB  | Test server, or very light use                            |
| <b>Recommended</b> | Standard_B2s_v2  | 2 vCPU · 8 GB  | Live system for 9–10 cars with room to grow               |
| <b>Growth</b>      | Standard_B4s_v2  | 4 vCPU · 16 GB | Many more cars or branches, or adding AI odometer reading |

**Other option:** the database can run on **Azure Database for PostgreSQL – Flexible Server** instead of inside the VM. Azure then handles database backups and updates, but the monthly cost is higher. For this size, the database inside the VM with daily backups is enough.

## 17.3 Why this size is enough

| Item                                    | Estimate                                                      |
|-----------------------------------------|---------------------------------------------------------------|
| <b>Cars</b>                             | 9–10                                                          |
| <b>Trips on a busy day</b>              | Up to about 50                                                |
| <b>People using it at the same time</b> | Fewer than 50                                                 |
| <b>Photo size after compression</b>     | About 300–500 KB, 2 photos per trip                           |
| <b>New photos</b>                       | About 40 MB a day → about 1.2 GB a month → about 15 GB a year |
| <b>Database size</b>                    | Well under 1 GB a year                                        |

With this load, a 2 vCPU / 8 GB VM will be mostly idle, and Blob Storage holds years of photos at low cost.

## 17.4 Cost note

The Azure retail price list (checked on 5 October 2026) shows Standard\_B2s\_v2 with Linux in Southeast Asia at about **US\$0.106 per hour, roughly US\$77 per month** for the VM when paid as you go. The disk, public IP,

Blob Storage and backup add a smaller amount on top. Reserving the VM for 1 or 3 years lowers the VM price. The final figure should be confirmed in the Azure Pricing Calculator with the company's Azure account.

## 17.5 Environments

- **Live (production):** the VM above.
- **Test (staging):** a smaller VM (Standard\_B2ls\_v2), or a separate test setup on the same VM, used to check every update before it goes live.
- Code is kept in Git. Each update is built and tested automatically, then deployed with one command.

## 18. Quality targets

| Area                | Target                                                                                         |
|---------------------|------------------------------------------------------------------------------------------------|
| <b>Devices</b>      | Android phones (Chrome) and iPhones (Safari) from recent years; Chrome or Edge on PC for admin |
| <b>Speed</b>        | Pages open in under 3 seconds on normal 4G; photo upload in a few seconds                      |
| <b>Data use</b>     | Photos compressed on the phone to about 300–500 KB                                             |
| <b>Uptime</b>       | 99.5% or better, with daily backups to restore quickly                                         |
| <b>Language</b>     | Bangla and English (to confirm)                                                                |
| <b>Time zone</b>    | Bangladesh time on all screens and reports                                                     |
| <b>Data keeping</b> | Trips and photos kept at least 1 year (to confirm); older photos can move to cheaper storage   |
| <b>Security</b>     | HTTPS only, hashed PINs and passwords, role-based access, full audit log                       |
| <b>Ease of use</b>  | A new driver can learn the flow in a few minutes with a one-page guide                         |

## 19. Delivery approach

We will build and roll out in this order:

1. Confirm the open points in Section 21 and sign off this plan.
2. Make screen designs (wireframes) and get feedback from 1–2 drivers and the transport team.
3. Build the backend: database, API, QR codes, photo upload, rules and audit log.
4. Build the driver and passenger screens.
5. Build the admin dashboard, reports and alerts.
6. Set up the Azure server, domain and SSL, and deploy a test version.
7. Test all flows and special cases on different phones and on weak networks.
8. Pilot with 1–2 cars and a few drivers; fix what we learn.
9. Print and fix QR stickers in all cars, give a short training to drivers and admins, and go live.
10. Support and improve based on feedback.



## 19.1 What we need from the company

- List of cars: car code, registration number, model, current km.
- List of drivers: employee ID, name, phone, licence number.
- Employee list for passengers: employee ID, name, department (Excel or from HR).
- Names of admins and viewers.
- Azure subscription access and a sub-domain for the app.
- Sticker printing (office printer with label paper, or a print shop).

## 20. Future add-ons

| Add-on                       | What it gives                                                                                                                     |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| <b>AI odometer reading</b>   | Reads the km from the dashboard photo automatically and flags it when it does not match the typed km. Our AI team can build this. |
| <b>Live GPS tracking</b>     | See the car's route on a map during the trip.                                                                                     |
| <b>Fuel log</b>              | Fuel fill-ups with receipt photo; km per litre per car.                                                                           |
| <b>Maintenance reminders</b> | Service by km (oil change, servicing) and expiry dates (fitness, tax token, insurance, driver licence).                           |
| <b>Notifications</b>         | SMS, WhatsApp or email to the passenger, driver or admin.                                                                         |
| <b>Car booking</b>           | Employees request a car in advance; admin assigns the car and driver.                                                             |
| <b>HR / ERP link</b>         | Employee list sync and cost sharing by department.                                                                                |

## 21. Points to confirm

| # | Question                                        | Our proposal                                                                                    |
|---|-------------------------------------------------|-------------------------------------------------------------------------------------------------|
| 1 | Trips without a passenger: who confirms them?   | Driver alone with start and end photos (and GPS). Shown separately in reports for admin review. |
| 2 | How do drivers sign in?                         | Employee ID + 4-digit PIN, once per phone; stays signed in.                                     |
| 3 | Is employee ID enough for the passenger?        | Yes for the first version (as asked). Add a PIN or SMS code later if misuse is seen.            |
| 4 | What if the passenger has no phone or internet? | Use any other phone; if not possible, admin approves later and the trip is marked for review.   |
| 5 | Can one trip have more than one passenger?      | One main passenger confirms; other names can be added as a note.                                |
| 6 | Do we save GPS at start and end?                | Yes, when the phone allows. No live tracking in the first version.                              |
| 7 | Screen language                                 | Bangla and English with a switch.                                                               |
| 8 | How long do we keep photos?                     | At least 1 year.                                                                                |
| 9 | Domain name for the app                         | A sub-domain like drive.<company domain>.                                                       |

| #  | Question                        | Our proposal                                         |
|----|---------------------------------|------------------------------------------------------|
| 10 | Who are the admins and viewers? | Transport manager(s) as admin; management as viewer. |